



UnionPay Integrated Circuit Card Specifications — Product Specifications

Part II Extended Purchase Specification Based on Contactless Low-value Payment Application



Version 2014

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Summary of Revisions

The change listed below is associated with the current version.

Description of Change	Where to look
Revised: Updated item “6986” from "It does not meet the condition for performing the command (it's not current EF)" to "Command not allowed (it is not the current EF; directly read or update the next record)".	A.1.5 A.2.5
Revised: Updated “DF82” to “DF7A”.	This document
Revised: Updated “DF83” to “DF7B”.	This document
Added: On the 2 nd step, added “If the CAPP transaction indicator bit = 1, and READ CAPP DATA command are not received or correctly answered before processing the GOP command, then the card will be processed normally.”	5.2.2
Added: Included the value of “DF61” if “DF61” in the 8th Byte of “9F69” if it is personalized. Otherwise “DF61” will be filled in with zero if it is not personalized. Terminal should check the 8th byte of “9F69” based on the FCI that contains “DF61”.	5.2.6 6.2.6
Added: Included a note after Appendix D Data Element.	Appendix D
Added: Appended “If R-MAC is supported, R-MAC will not be returned when UPDATE CAPP DATE CACHE returns an error.” in A.1.5.	A.2.5
Removed: If the card support SM2 algorithm, the returned information must include SM2-request indicator (tag DF69) and PDOL of terminal country code (tag 9F1A).	5.2.1

1 Application Scope

This book applies to all UPI participants.

2 References

UnionPay Integrated Circuit (IC) Card Specifications (All parts)

3 Terms and Definitions

This Standard utilizes the following terms and definitions:

3.1 CAPP records

Specified file records of extended application, including comprehensive application cyclic file and comprehensive application specified file

3.2 Command

A message that the terminal sent to the card. This message initiate an operation or request for a response.

3.3 Comprehensive application specified file

A file used to store specific industry application information. It's usually a record-structured file of variable length.

3.4 Comprehensive application cyclic file

Comprehensive application cyclic file is loop-structured file used to store log data. The "UPDATE CAPP DATACACHE" command only update the first record at each transaction..

3.5 Cryptogram

The calculated result of encryption operation.

3.6 Deposit deduction

Deposit deduction is a mechanism in which the issuer gives the cardholder a certain deposit amount. And this amount is not accounted for the available electronic cash balance. But when the available electronic cash balance is not sufficient to complete a transaction, the cardholder can choose to use the deposit amount to finish the transaction

3.7 electronic cash (EC)

Small-amount payment instrument developed based on debit/credit application.

3.8 Electronic cash balance

A counter to keep record of how much is available for offline purchase.

3.9 Electronic cash funds limit

The upper limit of electronic cash balance available for offline purchase.

3.10 Identify number

A number used to distinguish industry's applications in different locations.

3.11 Proximity payment systems environment

A list consisted of the supported application identifier, application tag and application preference indicator. And the list can also be accessed via contactless interface. It also includes entrance to all the directories, which is contained in FCI, in response to the “SELECT PPSE (“2PAY.SYS.DDF01”) ” command.

3.12 Load

A method to increase the electronic cash balance. There are several ways to implement load transaction. Money can be transferred to the electronic cash from the master account of the card, cash, or the account of other cards. But electronic cash balance after load transaction is still bounded by the electronic cash funds limit

3.13 Response

The message returned to the terminal after the IC cards finish the processing of received command

3.14 Offline pre-authorization

An estimated amount is sent to the card alongside the other parts of the transaction command. The card approves the transaction after risk analysis and balance check. And then the estimated amount will be frozen in the current electronic cash balance.

3.15 Offline pre-authorization completion

The transaction command is sent to the card before the pre-authorization expiry date. And then the card will approve the transaction after risk analysis and balance check. And the corresponding amount is deducted from the electronic cash balance.

3.16 Script

One or a series of commands sent from the issuer to the cards.

3.17 Section purchase

This is a variation of standard qUICS transaction flow chart to satisfy the need for deduction by segment or deduction by time in the quick offline low-value payment application.

3.18 Terminal

Equipment installed in the sales point to complete financial transactions alongside IC cards. It should include interface part or other parts (for example other part to communicate with the host)

4 Symbols and Acronyms

The following table of symbols and acronyms is applicable to this Specification:

AAC	Application Authentication Cryptogram
AC	Application Cryptogram
AFL	Application File Locator
AID	Application Identifier
AIP	Application Interchange Profile
ARQC	Authorization Request Cryptogram
ATC	Application Transaction Counter
CAPKI	CA Public Key Index
CAPP	Extended application
CDA	Complex Dynamic Data Authentication / Application Cryptogram Generation
CDOL	Card Risk Management Data Object List
CLA	Class Byte of the Command Message
CVM	Cardholder Verification Method. The Specifications support two cardholder authentication methods: Online PIN and signature
CVR	Card verification Results
CVN	Card Verification Number
dCVN	Dynamic Card Verification Number is a dynamic signature generated by the card application. Dynamic CVN replaces static CVN in equivalent data of magnetic track 2
DDA	Dynamic Data Authentication, prevents false card in offline authentication, card generates RSA signature for specific data in transaction which could be used in terminal verification.

DDF	Directory Definition File
DDOL	Dynamic Data Authentication Data Object List
DES	Data Encryption Standard
DKI	Derivation Key Index
Dynamic Signature	Dynamic signature is a signature generated by the card from the dynamic data of a card and a terminal, and the terminal will prove the authenticity of the card by verifying the signature. Dynamic application data (Tag '9F4B ') of signatures shall be referred when using
End Sentinel	It is the indicator in the end position of magnetic track 1 and magnetic track 2 of the magnetic stripe. It will be followed by a longitudinal redundancy check code - see "LRC". "
FCI	File Control Information, responded and returned by card when reader or terminal selects card application (using SELECT command)
fDDA	Fast DDA, a fast DDA in line with EMV and UICSUICS definition of norms. Used in qUICS transactions, allowing reader to give READ RECORD command to obtain dynamic data authentication related data from card and execute DDA calculation after card leaves induction zone.
GPO	GET PROCESSING OPTIONS
Hex	Hexadecimal
IAC	Issuer Action Code. In case it's determined that an offline transaction is rejected (IAC Denial), then it will be sent online (IAC Online); if online is infeasible, then return to offline rejection (IAC Default) conditions
IC	IC (Integrated Circuit)
ICC	IC card (Integrated Circuit Card)
ICVN	Alternative CVN, used as the mirror of equivalent data of personalization in magnetic track 2 of the chip
Lc	In commands of case 3 or case 4, it is the exact length for data field of commands sent from the terminal application layer (TAL)

IDD	Issuer Discretionary Data
Le	In commands of case 2 or case 4, it is the expected maximum length of data responded by the terminal application layer (TAL)
LRC	Longitudinal Redundancy Check. It's used to validate the data to ensure that the data will not be lost when being read from the physical magnetic stripe
ISO	International Standards Organization
LV	Low-value payment options of contactless qUICS
MAC	Message Authentication Code
MDK	Master key (Master DEA Key), is a double length DES key used to derive a unique key for the card (for online card authentication)
MSI	Magnetic Stripe Image, can not be used for contactless magnetic stripe technology
MSD	Magnetic Stripe Data refers to the contactless payment using equivalent data of magnetic track 2. Magnetic track 1 can be built within the equivalent data of magnetic track 2 and self-defined data of magnetic track 1 and is used for contactless transactions or online message
MSD Path	When a transaction is done through a contactless interface, MSD path is an application path adopted by the Card, to make the Card action meet the MSD requirements. The path will be used in case that the Reader supports MSD, the Card supports MSD, and qUICS is not supported.
P1	Parameter 1
P2	Parameter 2
PAN	Primary Account Number
UICSUICS	UICSUICS debit/credit application. UICSUICS in the Specifications refers to application supporting the contact interface and in full compliance with EMV 4.1 requirements.
PDOL	Processing Options Data Object List A list of terminal data objects required by card.

PIN	Personal Identification Number
PIX	Proprietary Application Identifier Extension
PPSE	Proximity Payment Systems Environment is a list of supported application identifications, application tags and application priority indicators, which can be accessed through a contactless interface. The list includes all the directory entries, and will be returned in FCI responded in SELECT PPSE ('2 PAY.SYS.DDF01') by the Card.
PSE	Payment System Environment
qUICS	Minimized UICS is to ensure a quick transaction can be done through a contactless interface. The Specifications describe the requirements of such transaction
RFU	Reserved for Future Use
RID	Registered Application Provider Identifier
RSA	A kind of asymmetric cryptographic algorithm used for encryption and authentication and developed by the Americans named Rivest, Shamir and Adleman
SDA	Static Data Authentication
SFI	Short File Identifier
SW2	Status Word 2 (Status Word Two)
TAL	Terminal Application Layer
TC	Transaction Certificate, application cryptogram generated when accepting transaction
TLV	Tag, Length, Value
UDK	Sub-key (Unique DEA Key) is the only double-length DES key of a Card derived from the master key, and is used for online certification of the Card

5 Transaction Process of Deduction by Segment

With the continuous spread of financial IC card applications, more and more industries are beginning to accept financial IC cards. Its security, fast, and offline payment features are more and more favored by the market.

Due to the special nature of related industries, new requirements have been raised for quick offline low-value payment applications: billing by time and billing by segment, where the Cardholder does not know the amount of purchase in advance, the amount of purchase calculated according to the environmental parameters before being deducted when the purchase action is completed. Typical scenarios include subway, highway (billing by segment) and parking meters (billing by time).

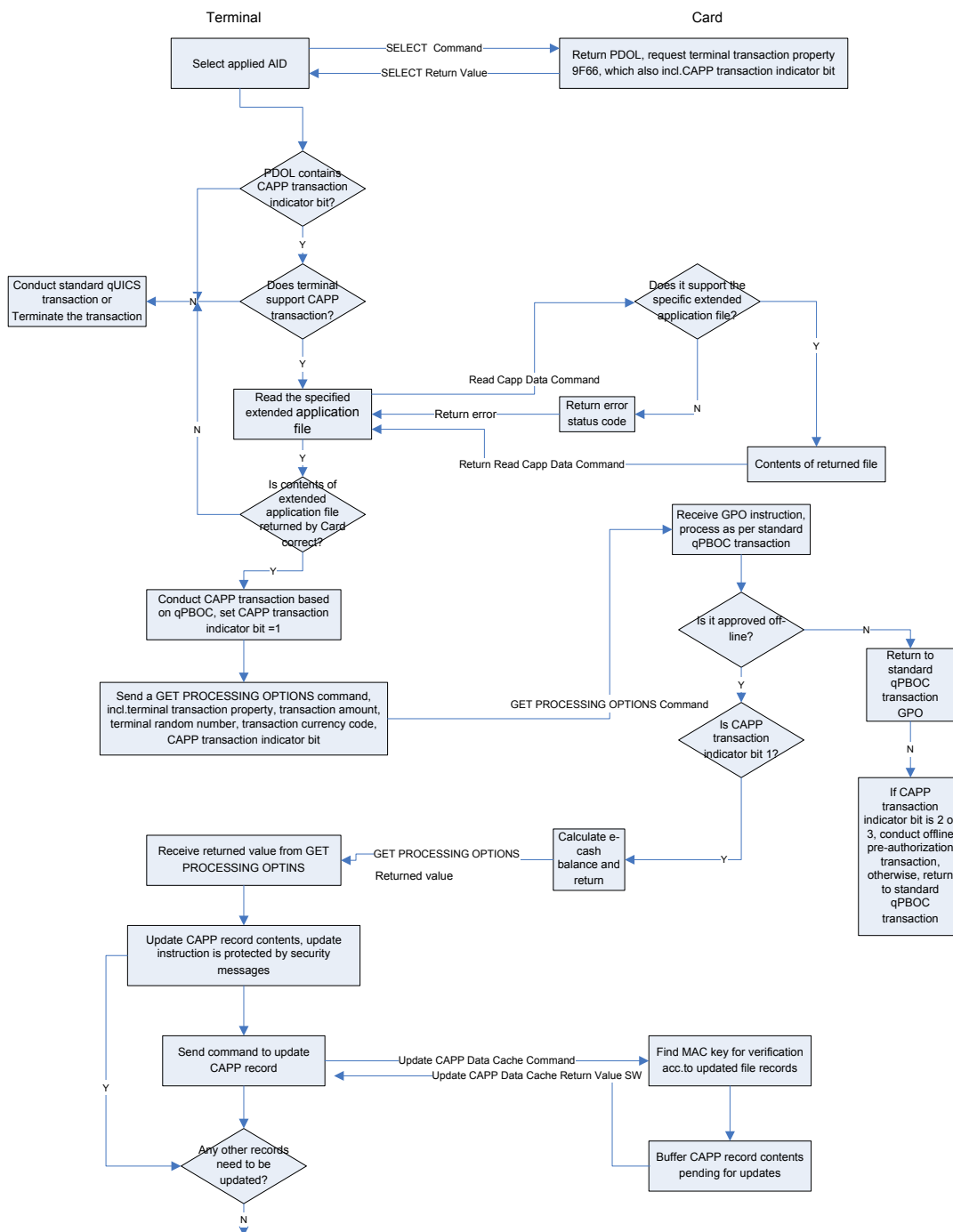
As the standard contactless quick payment cannot meet the new demands of these industries, adjustments and extensions are required for the existing procedures to accommodate the characteristics of these scenarios.

Based on the original standard qUICS transaction process, UPDATE CAPP DATA CACHE command is added between the GPO command processing and READ RECORD command processing to update the extended application data.

Deduction of GPO command and update of UPDATE CAPP DATA CACHE command must be performed simultaneously, and updated simultaneously when READ RECORD command successfully reads the last record of AFL.

In deduction by segment transaction mode, issuers may choose to select the deposit deduction function, which needs to add the two tags, i.e. deduction limit of deduction by segment (DF62) and the deducted amount of deduction by segment (DF63) during personalization.

5.1 Flow Chart for Deduction by Segment Transactions



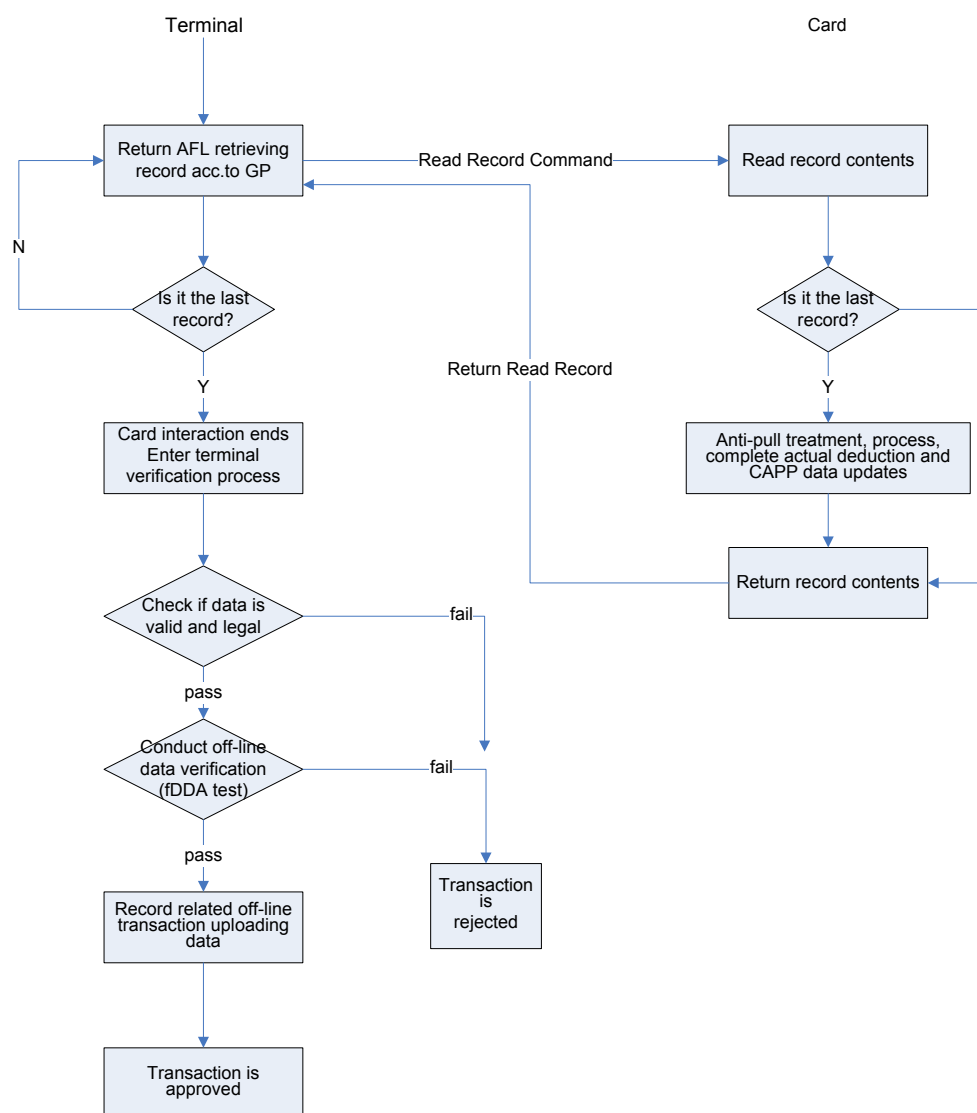


Figure 5-1 Flow Chart for Deduction by Segment Transactions

5.2 Description of Deduction by Segment Transaction Process

The following is the deduction by segment transaction process based on contactless low-value payment:

5.2.1 Selection of Applications

The terminal will send SELECT PPSE command to select PPSE following qUICS. According the application information and AID returned from the card. The terminal will send a SELECT command to select the application, and the card will return File Control Information (FCI). FCI also includes extension-function indicator (tag DF61). For example, the first bit of the first byte of DF61 is set as “1”, it indicates the cards support deduction by segment. If the eighth bit of the first byte is set as “1”, it indicates the card support R-MAC protection of the extended application records.

If PDOL data required by FCI data contains CAPP transaction indicator bit, the terminal will process in order as follows:

- Determine whether the terminal supports the extended application: if yes, continue the follow-up process; if no, CAPP transaction will indicate location "0", and carry out the contactless IC Card transaction or terminate the transaction as required;
- The terminal will send a Read Capp Data command to read the specified variable length record CAPP file, or send a Read Record command to read the specified cycle record CAPP file, to determine whether the Card supports the specific extended application;
- If the specified CAPP file exists in the card, the card will return the file contents, the terminal will confirm that the card supports the specific extended application, CAPP transaction will indicate location "1", and then continue the follow-up processing;
- If the specified CAPP file does not exist in the card, the card will return an error status code, indicating that the card does not support the specific extended application, the terminal will indicate CAPP transaction location "0", and carry out contactless IC Card transaction or terminate the transaction as required.

Through the deduction by segment identifier (DF61) = 0x01 in the issuer discretionary data (BF0C) of the File Control Information (FCI), the terminal will indicate that the card supports only the deduction by segment application.

For a terminal supporting deposit deduction transaction, it may get electronic cash balance (9F79) according to actual needs. When the acquired electronic cash balance (9F79) is greater than or equal to 0, the terminal can select to continue or terminate the transaction according to business needs.

5.2.2 Application Initialization

The terminal will send a GPO command to the card, and data in the command will be organized according to the returned PDOL data when the application is selected, and it is required to contain a CAPP transaction indicator bit.

When a GPO command is received, the card will be process in the following order:

- Process according to the standard qUICS transaction, and determine whether to approve offline the transaction : If yes, continue the follow-up process; if no, handle the GPO command according to the online/ rejection transaction process of standard qUICS;
- If the CAPP transaction indicator bit = 1, enter the deduction by segment transaction process; If the CAPP transaction indicator bit = 1, and READ CAPP DATA command are not received or correctly answered before processing the GOP command, then the card will be processed normally.

- If the card supports the deduction by segment and deposit deduction functions, the current actually available cash balance = electronic cash balance (9F79) + deduction limit of deduction by segment (DF62) – deducted amount of deduction by segment (DF63);
- If currently available electronic cash balance is less than current transaction amount, go to the standard qUICS process, and judge whether to refuse the transaction or to request on-line;
- If current actually available electronic cash balance is greater than or equal to current transaction amount, then current actually available electronic cash balance will replace the electronic cash balance (9F79) for low-value-payment check and other related operations (except for prepaid processing, which still uses electronic cash balances (9F79) as the basis for determination);
- If the CAPP transaction indicator bit = 2 or 3, then go to the offline pre-authorization transaction process, for details, please refer to 6.2;
- If the CAPP transaction indicator bit is of other values, then go to the standard qUICS transaction process.

5.2.3 Process of Deduction by Segment

During deduction by segment transaction, the terminal will perform the following processing:

- Organize and update the contents recorded in CAPP and the terminal will conduct security protection to the UPDATE CAPP DATA CACHE command by calculating corresponding MAC with the security module.
- Send an UPDATE CAPP DATA CACHE command. It's allowed to send multiple UPDATE CAPP DATA CACHE commands according to the actual application.

The card will perform the following processing:

- According to the file records indicated by the UPDATE CAPP DATA CACHE command, find corresponding protection key, calculate and verify the security message.
- Once the security message authentication is successful, CAPP application data will be cached, and written into the Card after the completion of the transaction.

5.2.4 Reading Card Data Content

- The terminal will send a READ RECORD command to the card according to the GPO returned AFL, and read corresponding record contents.

- After the last record is successfully read, the card will simultaneously complete low-value payment deduction and actually update CAPP records, and the transaction is completed normally.
- If the card supports the deposit deduction function, and the electronic cash balance (9F79) is less than the current transaction amount, then a deposit deduction will be carried out. Then after the last record is successfully read, the deducted amount of the deduction by segment after the transaction (DF63) = the deducted amount of the deduction by segment before the transaction (DF63) + transaction amount - electronic cash balance before transaction (9F79), while the electronic cash balance after the transaction (9F79) is set to zero; the transaction is completed.

5.2.5 Terminal Action Analysis

Please refer to the standard QUICS specifications.

5.2.6 Completion Processing

The terminal performs transaction completion steps (i.e., Terminal Authentication Process) and determines the result of transaction processing (transaction rejection or transaction approval). It includes the following steps:

- Check the validity and legitimacy of all related data;
- Offline data authentication, i.e. fDDA verification;

If the card's fDDA version number is "01", then before generating the dynamic signature, the card shall dynamically fill the 8th byte of the card verification related data (9F69) with the value of the card top-up function indicator (DF61) then making dynamic signature calculations. After fDDA verification is successful, the terminal shall compare the 8th byte of the card verification related data (9F69) with the card top-up function indicator (DF61) in the FCI data returned by the card upon selecting an application, if there is any mismatch, then it shall indicate that the translation failed.

Fill the 8th byte of 9F69 with value of 'DF61' if 'DF61' is personalized and fill with zero if 'DF61' is not personalized. Terminal should check the 8th byte of '9F69' based on if FCI contains 'DF61'.

5.2.7 Support Special Processing of Deduction by Segment and Deposit Deduction Functions

1. Loading operation

The loading procedure of the Issuer in background system should be consistent with the existing procedure.

When the card receives a script command regarding the modifying balance sent by the Issuer, First of all, the card does a subtraction to get the difference between the amount in command "PUT DATA 9F79" and the deducted amount (DF63) after

deduction by segment transaction. If the difference is greater than the electronic card balance limit, the card will return status "6A80" as a response to "PUT DATA" command

—— If current electronic cash balance (9F79) is equal to 0:

When the amount specified in the modified balance script is greater than the deducted amount in the deduction by segment (DF63), the loaded electronic cash balance (9F79) = the amount specified in the modified balance script - the deducted amount in the deduction by segment (DF63). Meanwhile, the deducted amount in the deduction by segment (DF63) shall be cleared to zero;

When the amount specified in the modified balance script is less than or equal to the deducted amount in the deduction by segment (DF63), the deducted amount in the deduction by segment (DF63) after loading = the deducted amount in the deduction by segment (DF63) before loading - the specified amount in the modified balance script, while the electronic cash balance (9F79) will remain unchanged;

—— If the current electronic cash balance (9F79) is greater than 0, then it will be handled according to the standard loading process.

2. Query Operation

—— Standard terminal can only support query on electronic cash balances (9F79) ;

—— Terminal supporting deduction by segment and deposit deduction functions can be separately used to query electronic cash balances (9F79), deduction limit of deduction by segment (DF62) and the deducted amount in the billing segment (DF63), while the query balance will be shown according to the actual business needs.

3. Operation for Updating Deduction Limit of Deduction by Segment

When a card receives a script command on modifying the deduction limit of deduction by segment sent by the Issuer, if the deduction limit of deduction by segment specified in the script on modifying the deduction limit of deduction by segment is less than the deducted amount in deduction by segment (DF63), then return to 6A80;

5.3 Description of the transaction process of second-currency deduction by segment

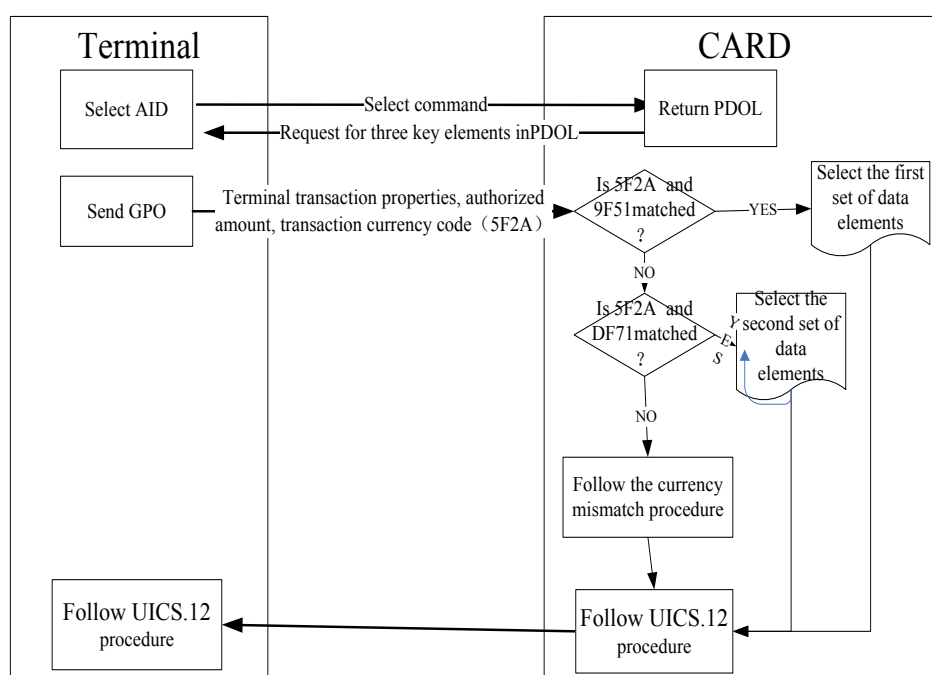
Second-currency electronic cash deduction by segment is a transaction based on second-currency qUICS. During the transaction, the card conduct risk management and balance update using a set of data elements or balance in accordance with the transaction currency code. The host of the issuer shall be able to differentiate the electronic cash accounts using the transaction currency code.

This section only describes the differences between the second-currency deduction by segment and standard deduction by segment. With respect to the other parts, please refer to the description of standard deduction by segment transaction process.

5.3.1 Initiate application

low-value

The card starts risk management after receiving GPO command. In the first step of card risk management (set the indicator of whether the currency is matched or not), the card will match the transaction currency code, and select the corresponding data elements. Illustrated as follows:



After receiving GPO command, the card will compare the transaction currency code (tag 5F2A) with the application currency code (tag 9F51). If they are matched, a set of the first-currency data elements will be used for the following processing. If not matched, the transaction currency code (tag 5F2A) will be compared with second-currency code (tag DF71). If they are matched, a set of the second-currency data elements will be used for the following processing. Otherwise, a set of the first-currency data elements will be used following the currency mismatch situation defined in qUICS.

After completing the card risk management, the card follows the rules below to respond to GPO command.

- If the available balance for offline purchase (tag 9F5D) is required in the response data, “9F5D” shall be calculated using either electronic cash balance (tag 9F79) or the second-currency electronic cash balance (tag

DF79), in accordance with the currency comparison result in the card risk management.

- If the electronic cash balance (tag 9F79) or the available balance for offline purchase (tag 9F5D) is required in the response data, the card shall calculate it using “9F79” or “DF79”, in accordance with the currency comparison result in the card risk management.
- the second-currency deduction by segment is only compliant with the option of “support small-amount balance check”, while it’s not applicable to either the option of “support small-amount balance check and CTTA check” or the option of “support small-amount balance check or CTTA check”. When the option of “support small-amount balance check and CTTA check” or the option of “support small-amount balance check or CTTA check” is used, the card shall disable the second-currency deduction by segment.

5.3.2 Read Application Data In the second-currency deduction by segment transaction, the card will carry on with the following procedure after processing GPO command

- The card shall return the second-currency electronic cash balance (tag DF79) in response to the terminal’s “GET DATA” command which is intended to get electronic cash balance (tag 9F79)
- The card shall return the second-currency electronic cash reset threshold value (tag DF76) in response to the terminal’s “GET DATA” command which is intended to get electronic cash reset threshold value (tag 9F6D)
- The card shall return the second-currency electronic cash funds limit (tag DF77) in response to the terminal’s “GET DATA” command which is intended to get electronic cash funds limit (tag 9F77)
- The card shall return the second-currency electronic cash single transaction limit (tag DF78) in response to the terminal’s “GET DATA” command which is intended to get electronic cash single transaction limit (tag 9F78)
- The card receives no GPO command, or the transaction currency code in the GPO command does not match with the second currency code (tag DF71) in the card’s application, the card will process the “GET DATA” command following qUICS procedure.

This aim of this design is to make the terminal able to identify and get the corresponding data elements in accordance with the transaction currency without making modification to the current transaction procedure of the terminal.

5.3.3 Special Handling in deduction by segment transaction

In the second-currency deduction by segment mode, the issuer could choose to enable the deposit deduction function. And two more tags DF7A (deduction limit of the second-currency deduction by segment) and DF7B (deducted amount of the second-currency deduction by segment) shall be added in the card personalization

template. In the meanwhile, special handling is also applied to the card supporting deposit deduction in parts of the second-currency deduction by segment or deduction by time transaction process.

1) Select Application

The terminal supporting the second-currency deduction by segment should get the second-currency electronic cash balance (DF79) before transaction. If the second-currency electronic cash balance (DF79) is greater than 0, the terminal can continue to start the transaction, if it's equal to 0 (that's the balance is 0. Or it has completed a deduction by segment transaction), the terminal can choose to continue or terminate the transaction according to its own logic.

2) Initiate Application

After receiving the GPO command, if the card also supports deduction by segment, the current available electronic cash balance can be calculated as follows:

The current available electronic cash balance = the second-currency electronic cash balance (DF79) + the second-currency deduction limit (DF7A) – the second-currency deducted amount (DF7B).

If the card does not support deduction by segment transaction, The current available electronic cash balance = the second-currency electronic cash balance (DF79) .

3) Read Card Data

After receiving the AFL from the card, the terminal sends “READ RECORD” command to the card. If the card supports deposit deduction and the second-currency electronic cash balance (DF79) is less than the transaction amount, the deposit deduction is conducted. And the rules below are followed.

The second-currency electronic cash deducted amount after transaction (DF7B) = the second-currency electronic cash deducted amount before transaction (DF7B) + transaction amount – the second-currency electronic cash balance (DF79). If the post-transaction deducted amount (DF79) is less than the second-currency deposit deduction limit, then the post-transaction deducted amount (DF79) will be updated after the last record is read, and the second-currency electronic cash balance (DF79) is set as zero. Otherwise, the transaction will fail.

4) The Load Operation

The load operation follows the current procedure at issuer.

After receiving “PUT DATA DF79” command, the card will carry out the following procedure:

The second-currency deducted amount (DF7B) is subtracted from the amount in “PUT DATA DF79” command. If the difference is greater than the

second-currency electronic cash funds limit (DF77), then a status word “6A80” will be returned in response to the “PUT DATA” command.

—— If the current second-currency electronic cash balance (DF79) is zero:

- If the amount in “PUT DATA DF79” command is greater than the second-currency deducted amount (DF7B), then the second-currency deducted amount (DF7B) is set as zero, and the second-currency electronic cash balance will be the difference between the amount in “PUT DATA DF79” command and the second-currency deducted amount (DF7B);
- If the amount in “PUT DATA DF79” command is less than the second-currency deducted amount (DF7B), the second-currency deducted amount (DF7B) will be set as the difference between the pre-transaction second-currency deducted amount (DF7B) and the amount in “PUT DATA DF79” command.

5) Inquiry Operation

- The standard terminal only supports the inquiry of the second-currency electronic cash balance (DF79) ;
- The terminal that supports deduction by segment transaction, also supports the inquiry of the second-currency electronic cash balance (DF79) , the inquiry of the second-currency deduction limit (DF7A) and also the inquiry of the second-currency deducted amount (DF7B). The corresponding balance will be displayed according to the business requirements.

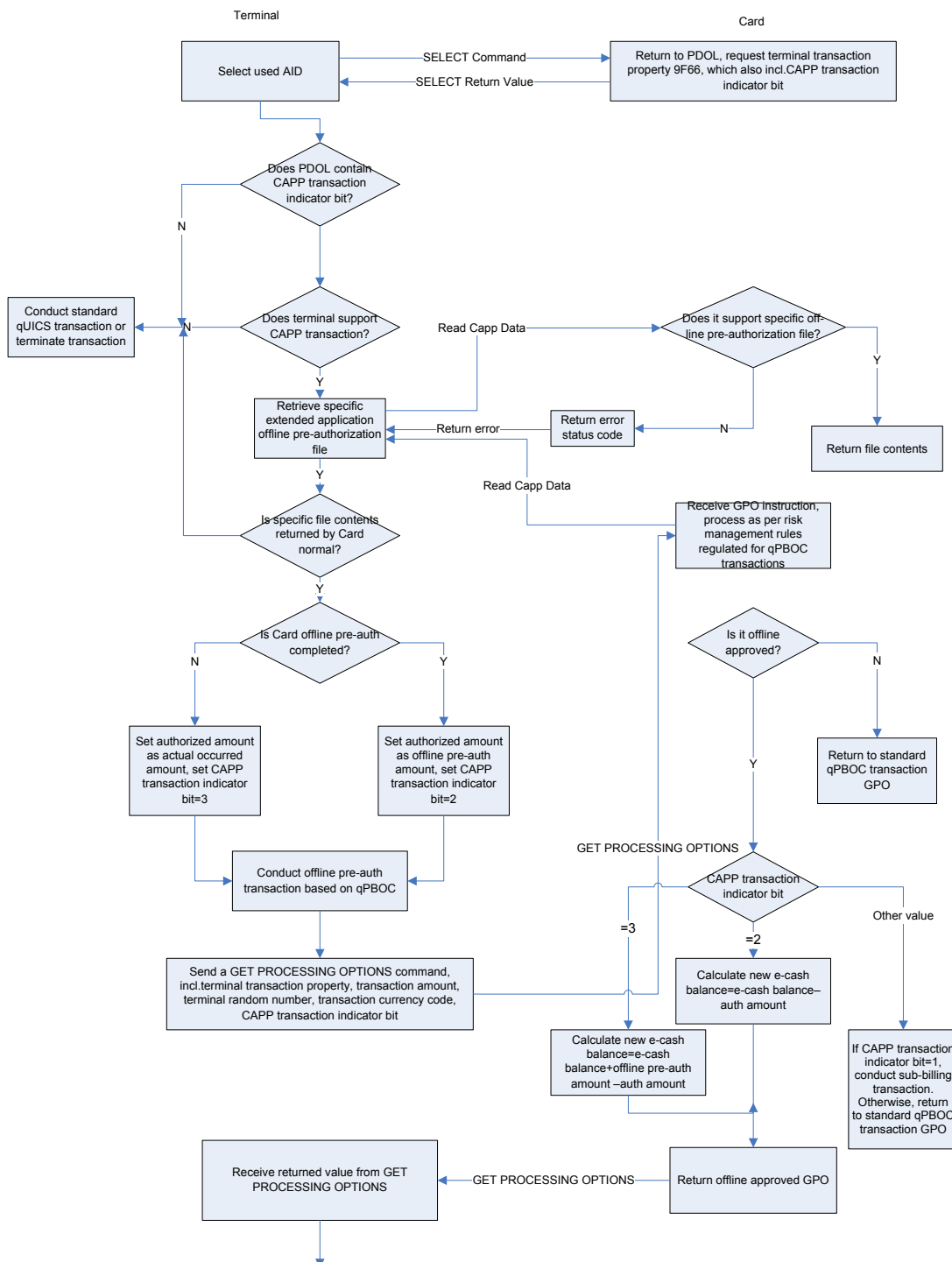
6) Updating the deduction limit operation

When the card receives the script to change the second-currency deduction limit (DF7A), the card will make comparison between the deduction limit in the updating script and the second-currency electronic cash deducted amount in the card. If the deduction limit in the updating script is less than the deducted amount in the card, the card will return “6A80”. Otherwise, the updating script will be executed. And the deduction limit in the card will be set as the deduction limit in the updating script.

6 Offline Pre-authorization Transaction Process

In order to meet the needs of offline pre-authorization scenario, based on the original deduction by segment transaction, CAPP transaction indicator bit and the application file shall be extended, and the following two types of transactions shall be newly added: offline pre-authorization transaction and offline pre-authorization completion transaction.

6.1 Flow Chart of Offline Pre-authorization Transaction



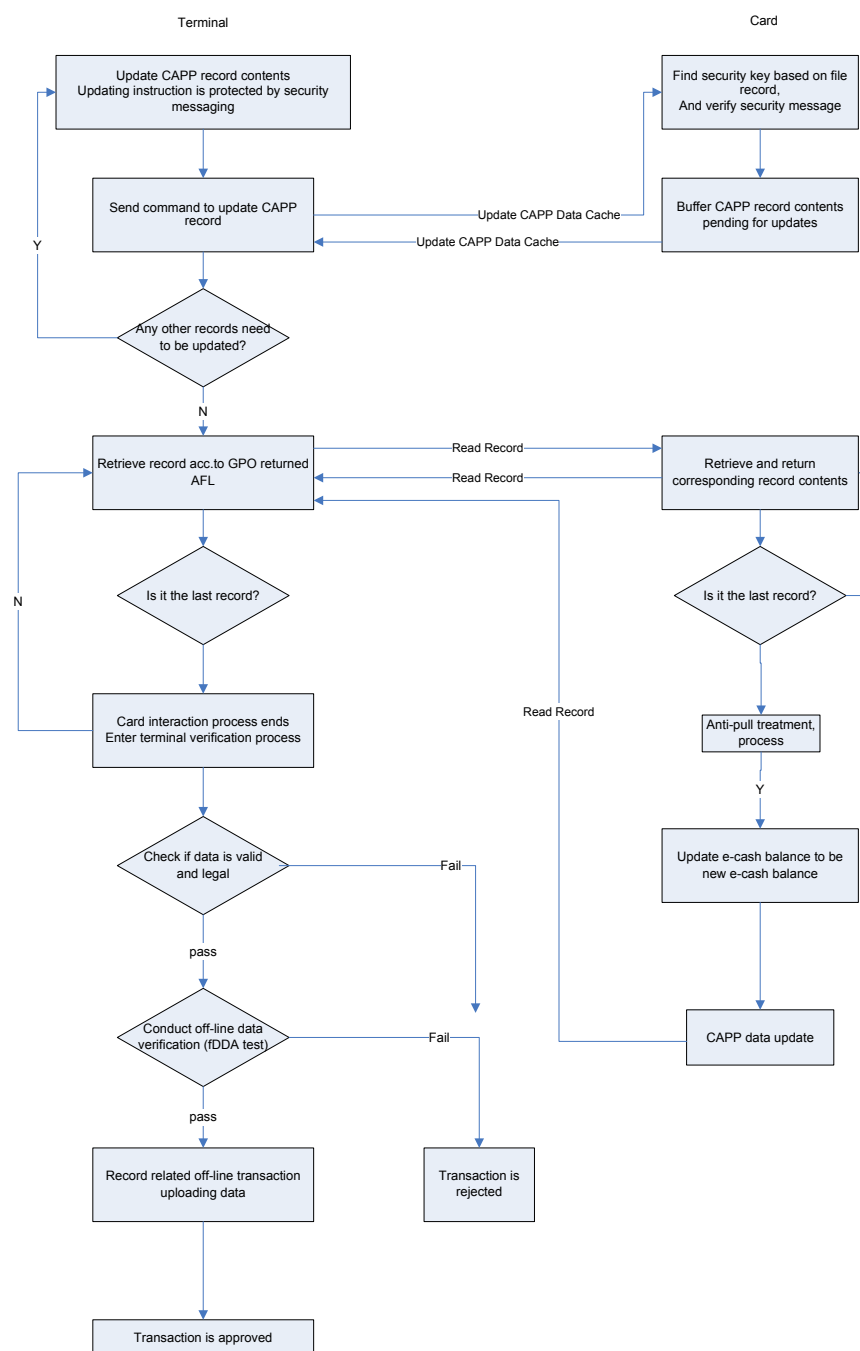


Figure 2 Offline Pre-authorization Transaction Flow Chart

6.2 Description of Offline Pre-authorization Transaction Process

The following is the transaction process for the offline pre-authorization transaction based on contactless low-value payment:

6.2.1 Selection of Applications

The terminal will send a SELECT command to select the application, and the card will return File Control Information (FCI).

If PDOL data required by FCI data contains CAPP transaction indicator bit, the terminal will process in order as follows:

- Determine whether the terminal supports offline pre-authorization applications: if yes, continue the follow-up process; otherwise the CAPP transaction indicator bit shall be set as "0", and carry out a standard qUICS transaction or terminate the transaction as required;
- The terminal sends a READ CAPP DATA command to read the specified variable length record extended application file, or send a Read Record command to read the specified cyclic record file, to determine whether the card supports the offline pre-authorization application or not;
- If there exists a specified offline pre-authorization application file in the card, the card will return the contents of the file, and the terminal will confirm that the card supports the specified offline pre-authorization application;
- If there does not exist a specified CAPP file in the card, the card will return an error status code, indicating that the card does not support the offline pre-authorization application, the terminal will set the CAPP transaction indicator bit as "0", and conduct a standard qUICS transaction or terminate the transaction as required;
- The terminal will determine according to the offline pre-authorization status in the read specified offline pre-authorization application data, the offline pre-authorization amount and the offline pre-authorization date or the validity (data in the offline pre-authorization application file is defined by the industry, but it is recommended to contain the offline pre-authorization status, the offline pre-authorization amount and the offline pre-authorization date or the validity), and the determination rules shall be set up by related industries according to their own actual needs. If it's determined that the transaction is an offline pre-authorization transaction, the terminal will set the CAPP transaction indicator bit as 2, set the authorization amount as the new offline pre-authorization amount, and conduct a new offline pre-authorization transaction; if it's determined that it's an offline pre-authorization completion transaction, the terminal will set the CAPP transaction indicator bit as 3, set the authorization amount as the actual transaction amount, and conduct an offline pre-authorization completion transaction;
- After completing the offline data authentication, the terminal will save all related transaction information for uploading. It's suggested that relevant information should contain the terminal ID Number and the merchant ID number of the terminal conducting the offline pre-authorization transaction, as well as related information of the local machine completing the offline pre-authorization transaction.

Otherwise, the terminal will perform the standard qUICS transaction or terminate the transaction as required.

The card requests the CAPP transaction indicator bit to the terminal through PDOL in the returned File Control Information (FCI), indicating that the selected application supports the deduction by segment transaction or the offline pre-authorization transaction based on contactless low-value payment.

6.2.2 Application Initialization

The terminal will send a GPO command to the card, and data in the command will be organized according to the returned PDOL data when the application is selected, and it is required to contain a CAPP transaction indicator bit. When a GPO command is received, the card will be processed in the following order:

- In case the transaction type is an offline pre-authorization completion transaction, the balance participating in card risk management should be the current balance plus the offline pre-authorization amount, and determine whether the transaction shall be approved offline according to the card risk management rules provided by qUICS: If yes, continue the follow-up process; if no, process the GPO command according to the online/rejection transaction process of standard qUICS;
- Determine whether the CAPP transaction indicator bit is contained in the GPO command data field, and set as "2" or "3". If CAPP transaction indicator bit is set as "2" or "3" and the card receive no READ CAPP DATA command or does not respond to READ CAPP DATA command correctly before the GPO command, the card should return "6985". If the CAPP transaction indicator bit is 2, the calculated new electronic cash balance = electronic cash balance - authorization amount, and the offline pre-authorization amount is recorded in the card for the offline pre-authorization completion transaction (currently it can support three offline pre-authorization transactions simultaneously, corresponding to three different internal offline pre-authorization amounts. In case the card receives a GPO command for the fourth offline pre-authorization transaction, the card will return '6971'); if the CAPP transaction indicator bit is 3, the calculated new electronic cash balance = electronic cash balance + offline pre-authorization amount – the pre-authorization amount;
- If the CAPP transaction indicator bit is 2, continuous offline pre-authorization transactions are not allowed to occur in the same application and in the same industry (i.e., the same ID under the same SFI) , and if the card receives a continuous offline pre-authorization transaction, it will return '6972';
- If the CAPP transaction indicator bit is 3, but the card can not find corresponding offline pre-authorization transaction, the card will return '6973';
- If the CAPP transaction indicator bit is 1, then it will enter the deduction by segment transaction process, for details please refer to 5.2;

- If CAPP is of other value, then it will enter into the standard qUICS transaction process;
- If the transaction is an offline pre-authorization transaction, the card will not return TC in case that offline transaction is approved; if it is an offline pre-authorization completion transaction, the card will return TC in case that the offline transaction is approved.

Table 6-1 GPO response necessities and conditional data based on the extended application transaction of contactless low-value payment

Tag	Mandatory (M) or conditional (C)	Data Elements
'82'	M	AIP
'94'	M	AFL
'9F36'	M	ATC
'9F26'	C If the transaction type is the offline pre-authorization completion transaction	Application Cryptogram
'9F10 '	M	Application data of the Issuer
'57'	C If the equivalent data of magnetic track 2 is not a part of static data pending for signature	Except equivalent data of magnetic track 2 is a part of static data pending for signature, it's necessary to have equivalent data of magnetic track 2
'5F34 '	C If it appears in the card	Application PAN Sequence Number
'9F4B '	C If it is supported fDDA and the ICC key length is less than or equal to 1024 bits	Signed Dynamic Application Data
'9F6C '	C If it appears in the card	Card Transaction Property

Tag	Mandatory (M) or conditional (C)	Data Elements
'9F5D '	C If it's allowed to show the offline amount and the ICC key length is less than or equal to 1024 bits	The offline purchase amount can be used unless the tag '9F5D' has been personalized and its value is '1', and the card shall not return this data element in the GPO response. Moreover, the Issuer should also give the card an additional process (byte 1 and bit 1) and personalize it to '1', indicating that the amount will be calculated and included in all contactless transactions. Tag '9F5D' will be personalized into '1', also indicating that the data element can be read by using the GET DATA command. The contents will be calculated according to the Issuer's instruction and the definition in the chapter and section regarding the additional process for the card (low-value-payment, low-value-payment and CTTA, low-value-payment or CTTA).
'5F20 '	O	Note to the cardholder's name: the cardholder's name is an required data element in the debit/credit application

6.2.3 Offline Pre-authorization Process

Once offline pre-authorization transaction is entered, the terminal will perform the following processing:

- In case the offline pre-authorization completion transaction takes place in the same terminal of the offline pre-authorization transaction, the terminal will use the transaction data generated by the offline pre-authorization completion transaction to cover the offline pre-authorization transaction data, and update the offline pre-authorization completion transaction record.
- For offline pre-authorization transactions, the terminal will update the offline pre-authorization amount and the date and time when the offline pre-authorization happens into CAPP records; for offline pre-authorization completion transaction, the terminal will use the offline pre-authorization amount and the date and time when the offline pre-authorization happens to cover the original CAPP record.
- Organize and update the contents recorded in CAPP, and the terminal will conduct security protection to the UPDATE CAPP DATA CACHE command by calculating corresponding MAC with the security module.
- Send an UPDATE CAPP DATA CACHE command, and buffer the CAPP record contents pending for update. It's allowed to send multiple UPDATE CAPP DATA CACHE commands according to the actual application.

- The card will perform the following processing:
- According to the file records indicated by the UPDATE CAPP DATA CACHE command, find corresponding protection key, calculate and verify the security message.
- Once the security message authentication is successful, CAPP application data will be cached, and written into the Card after the completion of the transaction.

6.2.4 Read Card Data Contents

- The terminal will send a READ RECORD command to the card according to the GPO returned AFL, and read corresponding record contents.
- The balance is not allowed to be updated by the card until the MAC in the UPDATE CAPP DATA CACHE command is successfully verified.
- When the last record is read, the card will check and ensure that the file updated by current UPDATE CAPP DATA CACHE must be consistent with the file of the last READ CAPP DATA (the same SFI and the same ID), after the successful update, the card will update the new electronic cash balance, and clear the corresponding pre-authorization amount at the same.
- The card will update the CAPP record contents. The above data update process is an atomic operation.

6.2.5 Terminal Action Analysis

Please refer to the standard QUICS specifications.

6.2.6 Completion Process

The terminal performs transaction completion steps (i.e., Terminal Authentication Process) and determines the result of transaction processing (transaction rejection or transaction approval). It includes the following steps:

- Check the validity and legitimacy of all related data;
- Offline data authentication, i.e. fDDA verification;
- After the terminal completes offline data verification, it stores all relevant transaction data for ease of upload. It is recommended that the relevant data shall include the terminal id and merchant id for the terminal where the pre-authorization offline transaction occurs, as well as the relevant data of the terminal where the pre-authorization offline transaction is completed.

If the card's fDDA version number is "01", then before generating the dynamic signature, the card shall dynamically fill the 8th byte of the card verification related data (9F69) with the value of the card top-up function indicator (DF61) then making dynamic signature calculations. After fDDA verification is successful, the terminal shall compare the 8th byte of the card verification related data (9F69) with the card top-up function indicator (DF61) in the FCI data returned by the card upon

selecting an application, if there is any mismatch, then it shall indicate that the translation failed.

Fill the 8th byte of 9F69 with value of 'DF61' if 'DF61' is personalized and fill with zero if 'DF61' is not personalized. Terminal should check the 8th byte of '9F69' based on if FCI contains 'DF61'.

6.2.7 Special Processing for Completion of Offline Pre-authorization Transaction

- During completion of offline pre-authorization transaction, if the offline pre-authorization completion amount is greater than or equal to the offline pre-authorization amount, then $\text{electronic cash balance} = \text{electronic cash balance} + \text{offline pre-authorization amount} - \text{offline pre-authorization completion amount}$;
- During completion of offline pre-authorization transaction, if the offline pre-authorization completion amount is less than the offline pre-authorization amount, and the deducted amount of deduction segment (DF63) is greater than 0, then $\text{offline pre-authorization remaining amount} = \text{offline pre-authorization amount} - \text{offline pre-authorization completion amount}$;
- If offline pre-authorization remaining amount is greater than the deducted amount of deduction segment (DF63), then zero out the deducted amount of deduction segment (DF63), concurrently set current electronic cash balance (9F79) = offline pre-authorization remaining amount - deducted amount of deduction segment (DF63); if the offline pre-authorization remaining amount is less than the deducted amount of deduction segment (DF63), then set the deducted amount of deduction segment (DF63) = deducted amount of deduction segment (DF63) - current offline pre-authorization remaining amount.

6.2.8 Load and Inquiry Operations for Offline Pre-authorization in Non-completion State

- Load Operation
Issuer backend load process remains consistent with existing process;
When the card receives the PUT DATA 9F79 command, the following processing should occur:
 - The card first determines the difference between the amount in the PUT DATA 9F79 command and the deducted amount of deduction by segment (DF63), checks whether it is greater than the electronic cash funds limit (9F77), and if so, then the card sends response code '6A80' for the PUT DATA command;
 - The card determines the amount in the PUT DATA 9F79 command, adds the not-yet-completed offline pre-authorization transaction amount (or total amount of multiple such transactions) on the card, then subtract the deducted

amount of deduction by segment (DF63). The card checks whether this is greater than the electronic cash funds limit (9F77), if so, then the card sends response code '6A76' for the PUT DATA command;

If the current electronic cash balance (9F79) is 0:

- If the amount in the PUT DATA 9F79 command is greater than the deducted amount of deduction by segment (DF63), then the card sets the deducted amount of deduction by segment (DF63) to zero. The card then sets the electronic cash balance (DF79) to the amount in the PUT DATA 9F79 command minus the deducted amount of deduction by segment (DF63);
 - If the amount in the PUT DATA 9F79 command is less than or equal to the deducted amount of deduction by segment (DF63), then the card sets the deducted amount of deduction by segment (DF63) to the current deducted amount of deduction by segment (DF63) minus the amount in the PUT DATA 9F79 command;
- If the current electronic cash balance (DF79) is greater than 0, then the card sets the electronic cash balance (DF79) to the amount in the PUT DATA 9F79 command. The electronic cash balance (7F79) or available offline purchasing amount (9F5D) returned by the card in response to GET DATA command or GPO command shall not include available balance to be deducted for not-yet-completed offline pre-authorization amount(s). The second-currency electronic cash balance (DF79) returned by the card in response to GET DATA command or GPO command also shall not include available balance to be deducted for not-yet-completed offline pre-authorization amount(s).

7 Single Deduction Discount Process Flow

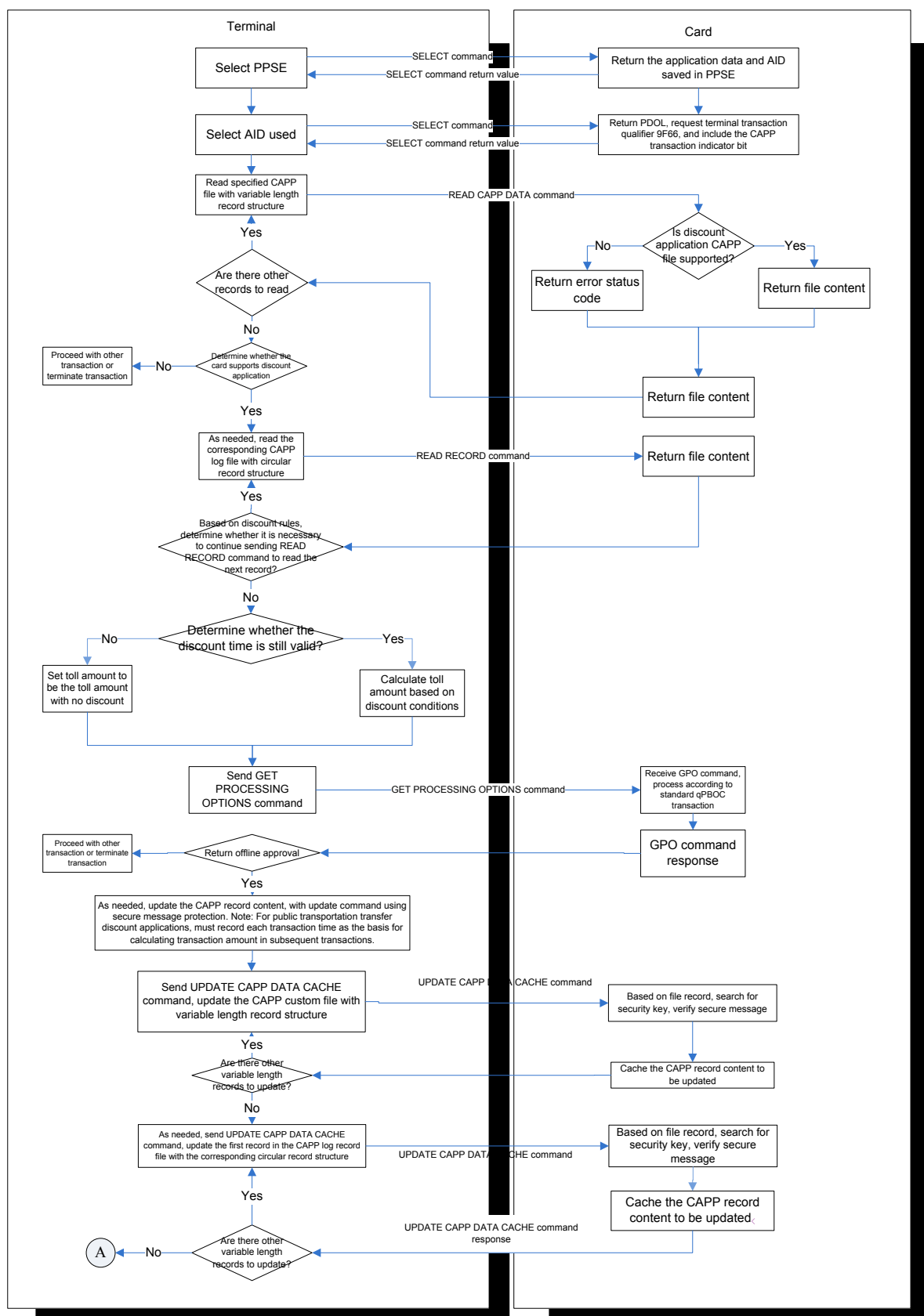
Single deduction discount refers to reading the extended application custom file data at transaction time to determine whether the card requires discount processing flow. Single deduction discounts are mostly applicable for bus transfer discounts, student cards, and senior citizen cards.

Description

Single deduction discount transactions have the following basic flow: read extended application custom file, determine whether the card supports discount application, if so, determine whether the discount period has expired; if not, calculate the purchase amount according to the discount rules, then continue with the other steps of the discount transaction.

Flow Chart

The single deduction discount transaction flow is shown in Figure 3 below



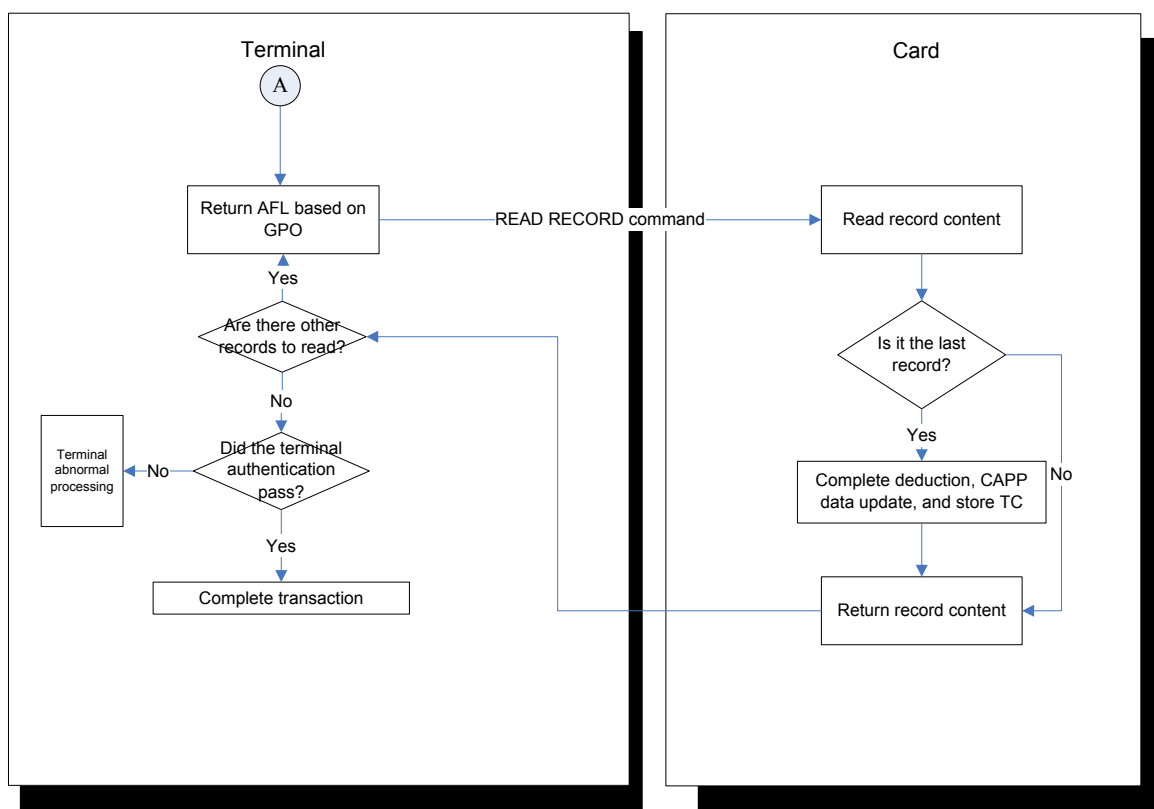


Figure 3 Single deduction discount transaction flow

Note: Single deduction discounts may involve checking the last or the last several transaction records (as in the case of bus transfers), so besides the CAPP variable length record file used by discount applications for general records, an additional CAPP log record file that uses circular record file structure may be added. By using the two files in conjunction, various discount plans could be realized.

Flow Description

When the cardholder uses contactless financial IC card for single deduction discount transactions in a discount application environment, the terminal will process it as follows:

- Terminal first selects and activates the card, then through response data determine whether the card supports extended application transactions based on contactless micro-payments.
- The terminal sends a READ CAPP RECORD command to check the variable length structure discount application CAPP custom file, to determine whether the card supports discount applications. If supported, the terminal sends one or more READ CAPP RECORD commands to read record content as needed from the circular record structure log record CAPP custom file: if the discount rules requires checking the recent several transactions, then the terminal must read the last several records from the circular file, as the basis

of the purchase amount calculation, otherwise only read the last log record as basis; if this does not meet discount conditions, then transaction amount is the non-discounted standard amount, otherwise, calculate purchase amount based on discount rules, then enter the deduction transaction flow.

—— Terminals that support discounts can calculate discount purchase amount by checking the recent several purchases. But avoid making the design overly complex, to limit impact on transaction speed.

Note: If the card supports extended application record R-MAC protection, then the terminal shall check and verify R-MAC. Specific methods are described in the fee deduction by segment and offline pre-authorization transaction flows.

8 Security Requirements

8.1 Description of Key

The qUICS application supporting extended application transactions will pre-set the proprietary file for the extended application specified in the specifications when personalizing (for details please refer to Appendix A). Each extended application file corresponds to one industry application, and the opening of each industry application will be protected by an access key in the way of security message. Access keys for these applications are derived from the same application access master key.

Table 8-1 Key Type Description

Key Type	Usage	Length	Management
Application opening master key	Application access keys generated through derivation are used to protect the extended application file	16 bytes	Issuer
Industry application governance key	It's used to control the authorization to modify data for industry application record	16 bytes	Industry partners

Note 1: The derived generation mechanism of application access key must ensure that the access key for each extended application file is unique.

Note 2: The generation and management of governance key for industry application shall be defined by the industry partner.

8.2 Security Mechanism

The terminal uses the APPEND RECORD command to add new application records in the extended application file of specified industry application, i.e. open a new industry application. The commands of using the APPEND RECORD command to add a new application in a variable-length record file, and the UPDATE CAPP DATA CACHE command to update the application file data are mandatory to contain security messages, so that the card is able to determine that the command is from a legitimate terminal.

The security message uses '00' || '00' || '00' || '00' || '00' || '00' || ATC as the initial vector to participate in the MAC calculation. For MAC calculation method, please refer to the description on message authentication codes in *UnionPay Integrated Circuit Card Specifications* 0025.7. Before sending the APPEND RECORD and UPDATE CAPP DATA CACHE command, the ATC can be obtained either by GET DATA, or through the returned value of GPO.

In APPEND RECORD command, the settings of governance keys for the industry applications are attached. After the opening of an industry application, the authorization to modify the application data of this industry application shall be controlled by the corresponding industry application governance key in the manner of security message. The terminal modifies industry application data with the UPDATE CAPP DATA CACHE command. The extended application supports the deletion function, i.e., the application validity identification shall be set to zero by the industry terminal when updating the industry data.

9 Personalization Requirements for Extended Application

UICS debit/credit applications supporting extended application transactions based on contactless low-value payment adopts the personalization methods of EMV card Personalization Specification (CPS) and the UICS/UICS 2.0 Debit/Credit Application Personalization Guide.

For cards only supporting deduction by segment applications, when the data is personalized, the Issuer should write a deduction by segment identifier (DF61) = 0x01 in the issuer discretionary data (BF0C) in the file control information (FCI) returned by SELECT AID.

For cards only supporting deduction by segment applications, in the opening process, the terminal should set the extended application identifier to 1 when adding a record to the proprietary file of extended application.

When personalization is used in PDOL of contactless interface, it shall contain the CAPP transaction indicator bit. For a contactless interface, if there is a CAPP transaction indicator bit in the PDOL returned by the card, it indicates that the card supports the comprehensive purchase transaction based on contactless low-value payment. If the terminal also supports the comprehensive purchase transaction, the GPO command will provide a CAPP transaction indicator bit, which will be set with a corresponding value; If the terminal does not support the comprehensive purchase transaction, the CAPP transaction indicator bit set as 0 will be provided to the card with the GPO command.

The issuer will create ahead of time a extended application file during personalization, and write in the initial key. Before using, it shall be opened by the cardholder on the industry devices, i.e., the industry will create a record corresponding to the file, and write in the read-write key of the industry to the record.

For cyclic record files, the files shall be created ahead of time by the Issuer in personalization process. If the industry needs to use the cyclic record file, a new record can be added with the APPEND RECORD command when opening the business by the cardholders, and then records in the file will be updated through UPDATE CAPP DATA CACHE. This file corresponds to one industry key.

Appendix A Proprietary File for Extended Application

A.1 READ CAPP DATA (read extended application data) Command

A.1.1 Definition and Scope

READ CAPP DATA command is used in the extended application transactions, where the terminal will retrieve information such as whether the card supports the extended application and information of last time extended application purchase transaction.

The terminal will determine the specified record of certain extended application file to be read through specifying the ID number of the extended application (generally the area code) and the industry type of extended application.

Once the card receives a READ CAPP DATA command, it will perform the following operations:

- Select the corresponding EF file according to SFI specified by P2. If the file does not exist, the card will return the status code '6A82' (file not found).
- If EF file is not a variable length record file, the card will return a status word '6981' (file type does not match).

A.1.2 Command Message

This command message is as follows:

Table 7-1 Command Message of READ CAPP DATA

Code	Value
CLA	'80'
INS	'B4'
P1	'00'
P2	Refer to Table 7-2
Lc	'02'
Data	ID #
Le	Not exist

The reference control parameter P2 in this command message is defined in the following table:

Table 7-2 Definition of the reference control parameter P2 in READ CAPP DATA command message

B8	B7	B6	B5	B4	B3	B2	B1	Meaning
0	0	0	0	0	-	-	-	RFU
X	X	X	X	X	-	-	-	SFI
1	1	1	1	1	-	-	-	RFU
-	-	-	-	-	0	0	0	Record appears in the first area code
-	-	-	-	-	0	0	1	Record appears in the next area code
-	-	-	-	-	X	X	X	RFU
Other values								RFU

A.1.3 Data field of the command message is an ID number of 2 bytes.

Data field of the command message is an ID number of 2 bytes.

A.1.4 Response Message Data field

Specify the record content of ID number.

A.1.5 Status Code of Response Message

Status code for successful performing of the command is '9000'.

Error status codes that may be returned by the IC card are as follows:

SW1	SW2	Meaning
'65'	'81'	Memory failure (modification failed)
'67'	'00'	Length error (Lc field is blank)
'69'	'81'	Command is incompatible with the file structure
'69'	'86'	Command not allowed (it is not the current EF; directly read or update the next record)
'6A'	'81'	This feature is not supported
'6A'	'82'	File not found

'6A'	'83'	Record not found
'6A'	'84'	Storage space of the file is not enough
'94'	'07'	The application is prohibited

A.2 UPDATE CAPP DATA CACHE (Update Data Cache) Command

A.2.1 Definition and Scope

UPDATE CAPP DATA CACHE command is used to update the application data cache in the extended application purchase transactions. Cached data will be used by the last READ RECORD command to rewrite related records in the proprietary files of the extended application. After the card receives the UPDATE CAPP DATA CACHE command, it will perform the following operations:

- Select the corresponding EF file according to SFI specified by P2. If the file does not exist, the card will return the status code '6A82' (file not found), and the terminal shall terminate the extended application purchase transaction.
- Check the condition for using the proprietary file of the extended application, if the previous command of this command is not a GPO command or is another UPDATE CAPP DATA CACHE command, it will return status code '6985' (the using conditions are not met). The terminal shall terminate this extended application purchase transaction.
- If the proprietary file of the extended application pending for updating is a variable length record file, according to the ID number in the command data field, check whether there is a record of the same ID number in the proprietary file of the extended application. If it does not exist, return a status code '6A83' (No Record found). The terminal shall terminate this extended application purchase transaction.
- Check whether the data field length of command is greater than the corresponding length recorded in the proprietary file of the extended application. If it's greater, return a status code '6A84' (file storage space is not enough). If it's less, return the state '6A80' (data field is not correct). The terminal shall terminate this extended application purchase transaction.

After passing the above-mentioned checks, the card shall temporarily save the SFI, record number and data field in the command. This command can't be used to update the corresponding data recorded in the proprietary file of the extended application.

An UPDATE CAPP DATA CACHE command is allowed to be performed for multiple times to update several records.

The proprietary file of the extended application can be of either variable-length record structure, or cyclic record structure. If it's of variable-length record structure, before using the UPDATE CAPP DATA CACHE command to update the extended application data, it must be ensured that there is a corresponding record in the file; if it's of cyclic record structure, each time when executing the command, it shall update the latest record and then use it in cycle.

In extended applications for contactless low-value payment, the command must utilize the security messaging mode.

A.2.2 Command Message

This command message is as follows:

Table 7-3 UPDATE CAPP DATA CACHE Command Message

Code	Value
CLA	'84'
INS	'DE'
P1	'00'
P2	Refer to Table 7-4
Lc	Length of follow-on data field
Data	For details please refer to the description
Le	Command does not exist

The reference control parameter P2 in this command message is defined in the following table:

Table 7-4 Definition of Reference Control Parameter P2 in the UPDATE CAPP DATA CACHE Command Message

B8	B7	B6	B5	B4	B3	B2	B1	Meaning
0	0	0	0	0	-	-	-	RFU
X	X	X	X	X	-	-	-	SFI
1	1	1	1	1	-	-	-	RFU
-	-	-	-	-	0	0	0	The record for the first appeared ID number (variable-length record file) or the latest record (cyclic record file)

B8	B7	B6	B5	B4	B3	B2	B1	Meaning
-	-	-	-	-	0	0	1	Record of the next appeared ID number (variable-length record files)
-	-	-	-	-	X	X	X	RFU
Other values								RFU

A.2.3 Command Message Data Field

Command message data field contains record content and security message.

If the current file is a variable-length record file, the record contents shall contain the ID number, record length, and other extended application information and data; if the current file is a cyclic record file, the command message data field shall contain the extended application data.

A.2.4 Response Message Data Field

Response message data field does not exist.

A.2.5 Response Message Status Code

Status code for successful performing of the command is '9000'.

Error status codes that may be returned by the IC card are as follows:

SW1	SW2	Meaning
'65'	'81'	Memory failure (modification failed)
'67'	'00'	Length error (Lc field is blank)
'69'	'81'	Command is incompatible with the file structure
'69'	'86'	Command not allowed (it is not the current EF; directly read or update the next record)
'6A'	'80'	Data field is not correct
'6A'	'81'	Function is not supported
'6A'	'82'	File not found
'6A'	'83'	Record not found
'6A'	'84'	Storage space of the file is not enough
'94'	'07'	The application is prohibited

If R-MAC is supported, R-MAC will not be returned when UPDATE CAPP
DATE CACHE returns an error.

A.3 APPEND RECORD Command

A.3.1 Definition and Scope

APPEND RECORD command is used to add extended application record to the extended application file when opening the extended application. It can be used to add records to the cyclic record file, also initialize the first record in the cyclic record file to the extended application.

After the card receives the APPEND RECORD command, it will perform the following processing:

- Determine whether the newly added record length exceeds the maximum length of file record, if it exceeds, the card will return the status word "6A80";
- Determine whether remaining space of the file is sufficient, if the space is insufficient, the card will return status word "6A84";

Through the above determinations, the card will allocate the record spaces according to the record data length of the command data field, and write new record data into the file.

A.3.2 Command Message

This command message is as follows:

Table 7-5 APPEND RECORD Command Message

Code	Value
CLA	'04'
INS	'E2'
P1	'0'
P2	Please refer to Table 7-6
Lc	Length of follow-on data field
Data	16-byte record modifying key (encrypted by the initial key) + newly added record content + MAC
Le	Does not exist

The reference control parameter P2 in this command message is defined in the following table:

Table 7-4 Definition of Reference Control Parameter P2 in APPEND RECORD Command Message

B8	B7	B6	B5	B4	B3	B2	B1	Meanings
0	0	0	0	0	-	-	-	RFU
X	X	X	X	X	-	-	-	SFI
1	1	1	1	1	-	-	-	RFU
-	-	-	-	-	X	X	X	RFU
Other values								RFU

A.3.3 Command Message Data Field

This command message is composed by 16-byte record modifying key, newly added record contents (extended application data) and MAC.

A.3.4 Response Message Data Field

Response message data field does not exist.

A.3.5 Response Message Status Code

Status code for successful performing of the command is '9000'. Error status codes that may be returned by the IC card are as follows:

SW1	SW2	Meaning
'65'	'81'	Memory failure (modification failed)
'67'	'00'	Length error (Lc field is blank)
'69'	'81'	Command is incompatible with the file structure
'69'	'86'	It does not meet the conditions for performing the command (is not current EF)
'6A'	'81'	Function is not supported
'6A'	'82'	File not found
'6A'	'83'	Record not found
'6A'	'84'	Storage space of the file is not enough
'94'	'07'	The application is prohibited

A.4 GET TRANS PROVE (Get Offline Transaction Application Cryptogram) Command

A.4.1 Definition and Scope

GET TRANS PROVE command is used to obtain TC (offline transaction application cryptogram) of extended application transaction corresponding to specific ATC (application transaction counter). The usage scenario is, in case the terminal cannot receive the response data of latest transaction command, it will re-power and send such command to obtain the TC of last failed transaction. If the response to the command succeeds, the terminal will determine that the last transaction is successful; otherwise, it will process according to transaction failed. This command can only obtain the TC of the latest extended application transaction that is the successfully completed by the card recently. If the latest transaction is a pre-authorization transaction, all the returned TC shall be zero.

A.4.2 Command Message

This command message is as follows:

Table 9-3 GET TRANS PROVE Command Message

Code	Value
CLA	'80'
INS	'5A'
P1	'00'
P2	'00'
Lc	'02'
Data	Transaction ATC specified by the terminal
Le	'08'

A.4.3 Command Message Data Field

Command message data field is composed by transaction ATC specified by the terminal.

A.4.4 Response Message Data Field

Response message data field returns the TC (8 bytes) corresponding to transaction ATC specified by the terminal.

A.4.5 Response Message Status Code

Status code for successfully performing of command is '9000'. Error status codes that may be returned by the IC card are shown in the following table:

SW1	SW2	Meaning
65	81	Memory failure
67	00	Length error
69	85	Usage conditions are not met
6D	00	Command does not exist
6E	00	Command type is not supported
94	06	Required TC is not available

Appendix B Proprietary File for Extended Application

B.1 Proprietary File for Extended Application

Proprietary file for extended application is usually of variable-length record structure, but according to the application requirements, such as in the transfer discount application mode, a cyclic record file for extended application, which is of cyclic record structure, is also needed to save the information as corresponding transfer records. (For the application examples in the specifications, all the proprietary files for extended application without special notes are of variable-length record structure). For details, refer to the table below:

If an proprietary file for extended application is of variable-length record structure, the proprietary file for extended application shall be created according to the information in below table.

Document Name	Variable-length proprietary file for extended application		Document Type :	Variable-length record file
Document Identifier	SFI (for recommended value, please refer to Appendix B)		File Size	Self-defined
Maximum length of a single record	Self-defined			
File authorization	Read	Open		
	Update	Protected		
Record Coding				
Byte	Data element			Length (bytes)
1-2	ID (area code)			2
3	Record length (length of follow-up data)			1
4	Application effective ID (0 indicates invalid; 1 indicates valid)			1
5	Extended application identifier (1 indicates that this application applies the deduction by segment; 2 indicates that the application applies offline pre-authorization purchase)			1
6	Application block flag (0 indicates that the application is not blocked; 1 indicates that the application is blocked)			1
7-n	Application Data			n-6

For specific definition of short file identifier of extended application, please refer to Appendix B.

B.2 Cyclic record file for extended application

Cyclic record file for extended application is of cyclic record structure. For each transaction, UPDATE CAPP DATACACHE command can only update the first record, and is only stored as logging data record. It's suggested to create a proprietary file for extended application according to the information in below table.

This document can also be used for other self-defined applications of the industry.

Document Name	cyclic record files for extended application		Document Type :	cyclic record files
Document Identifier	SFI=0x1E		File Size	Self-defined
File authorization	Read	Open		
	Update		Protected	
Record Coding				
Byte	Data element			Length (bytes)
1-n	Extended application data			n

The application supporting transfer discount should record the details of this transaction in the cyclic record file for extended application. For transfer discount, it is required to read the contents in cyclic record file as the basis for discount transfer.

Appendix C Recommended Definitions for Extended Applications

According to the actual application needs of current several common extended applications, and combined with the realization modes of applications in the specifications, the recommended definitions of the short file identifier (SFI) and the access key of extended application are as below. For details, please refer to table below

Extended application type	Extended application file SFI	Access key
Ordinary subway application	0x15	Pre-set
Ordinary bus application	0x16	Pre-set
Non-stop fare charging for Highway	0x17	Pre-set
Parking fare charging for meter application (internal balance)	0x18	Pre-set
Railway (High Speed Rail) application (internal balance)	0x19	Pre-set
General bus daily ticket/monthly ticket application	0x1A	Pre-set
General subway daily ticket/monthly ticket application	0x1B	Pre-set
Other extended applications (internal balance)	0x1C	Pre-set

Appendix D Data Element

Those data elements referred to in this Specification, but not defined or modified in UICSUICS specifications will be defined in this Appendix.

Name	Format Tag Length	Requirements	Description	Retrieve	Value
CAPP transaction indicator bit	F: b8 T: DF60 L: 1	Conditions If the card supports a CAPP comprehensive transaction, it is required to specify the data in PDOL	Specify the CAPP transaction type supported by the terminal	N/A	0: Indicates that the terminal does not support the extended application 1: Indicates that the terminal supports deduction by segment transactions 2: Indicates that the terminal supports offline pre-authorization transactions 3: Indicates that the terminal supports offline pre-authorization completion transactions
Deduction by Segment Application Identifier	F: b8 T: DF61 L: 1	If the card only supports deduction by segment applications, the Issuer will conduct the personalization in BF0C	It's used to distinguish the capability of the card in supporting extended applications	Returned in the Issuer discretionary data BF0C in the file control information (FCI)	1: Indicates that the card only supports deduction by segment applications 2: Indicates that the card supports the extended application

Deduction limit of deduction by segment for electronic cash	F: n12 T: DF62 L: 6 Format: cn	If the card supports the deduction by segment deduction function, it will support the Get Data and Put Data commands.	It indicates the maximum deductible amount for deduction by segment transactions	It can be returned through the Get Data command	
The deducted amount of electronic cash deduction by segment	F: n12 T: DF63 L: 6 Format: cn	If the card supports the deduction function of deduction by segment, it will support the Get Data command, but not the Put Data command.	Indicates the current deducted amount of the card	It can be returned through the Get Data command	

Note1: If 'DF61' indicator doesn't support extended application transaction, but terminal request an extended application transaction in GPO. To be compatible with terminal, card will still support extended application transaction.

Note2: If 'DF61' indicator doesn't support R-MAC, but when terminal request R-MAC, card is allowed to return R-MAC to complete transaction.

Appendix E (Informative Annex)

Example deduction by segment transaction applications

This appendix provides examples based on contactless micro-payment for deduction by segment transactions in specific application environments, describing the actual application modes for deduction by segment transactions.

Note: The following example description has the pre-condition that the terminal supports the specified deduction by segment transaction application.

E.1 Subway/Railway/Express-way Tolls/Parking Meter Applications

Subway/railway/expressway toll applications utilize fee billing by segment model; parking meter applications utilize fee billing by time, but all the applications above use the same transaction flow. This section will go in-depth into description of a specific contactless micro-payment based deduction by segment application's transaction flow, using a subway billing application as the example. The entire transaction is divided into a gate-entry transaction (or toll zone entry transaction) and a gate-exit transaction (or toll zone exit transaction). Regarding the CVM settings on the terminal, implement in line with the electronic cash requirements.

E.1.1 Subway Billing Application

E.1.1.1 Gate-entry transaction flow

Description

The basic flow for the gate-entry transaction is: Select the PPSE payment environment, then select the qUICS application, read the extended application custom file, then determine whether the last transaction was completed successfully. If the last transaction was completed successfully, then engage in purchasing and update the file; otherwise return an error prompt, to show that the cardholder is not permitted to enter the toll zone.

Terminal can also engage in pre-processing based on actual circumstances, for example can preemptively obtain the card balance, to determine whether to permit the cardholder to enter the station.

Flow Chart

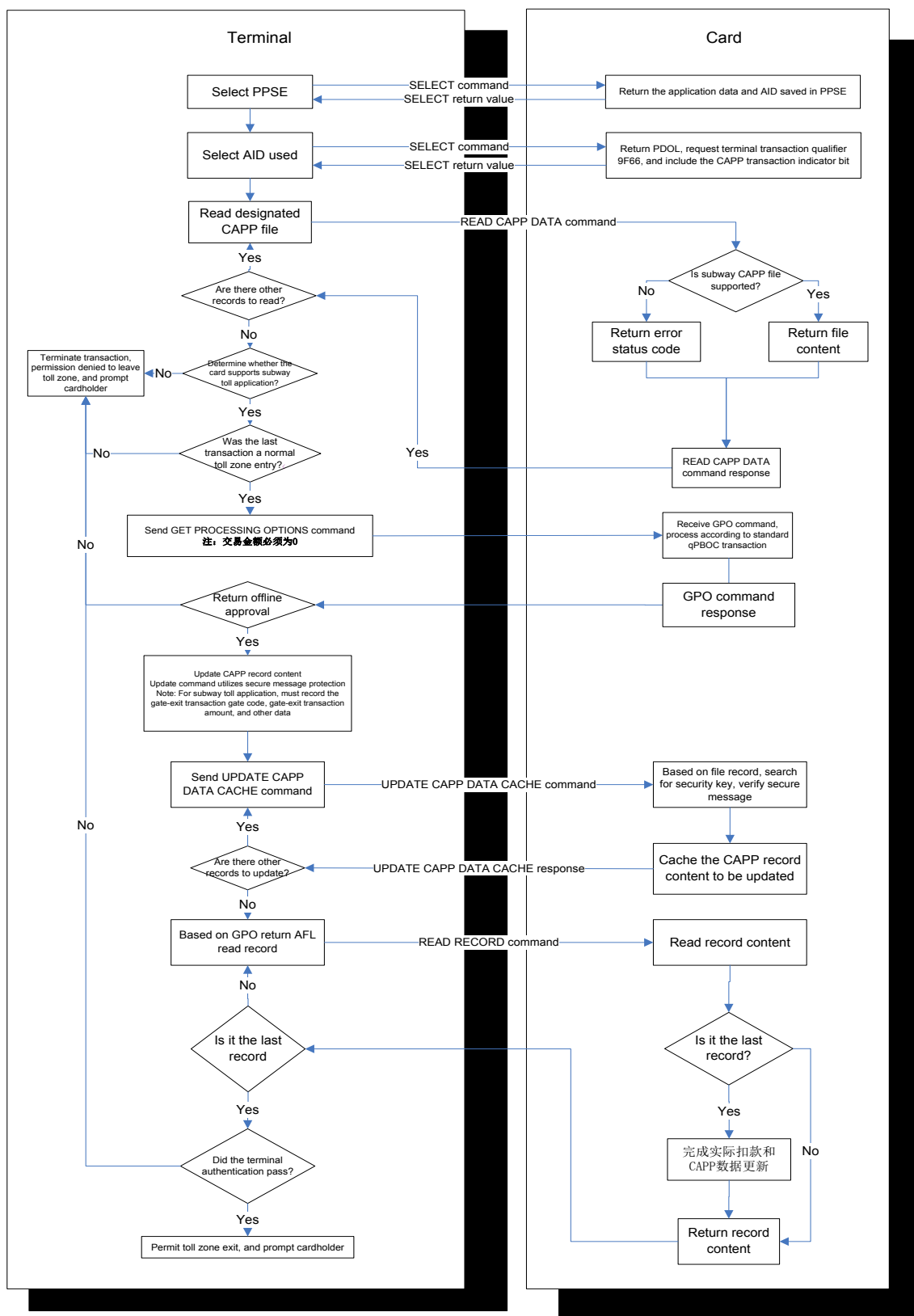


Figure E.1 Gate-entry transaction flow for subway toll application

Flow Description

When the cardholder uses contactless financial IC card for gate-entry transactions in a subway toll application environment, the terminal will process it as follows:

- Terminal first selects and activates the card, then through response data determine whether the card supports extended application transactions based on contactless micro-payments through AID selection.
- Terminal sends READ CAPP RECORD command inquiry, to determine whether the card supports subway toll application. If supported, the terminal shall read this specified custom data, and process according to the data, for determining whether the last transaction was leaving the toll zone, etc. If the processing result is that the gate-entry transaction is not permitted, then the terminal shall notify the cardholder. If the processing result permits the gate-entry transaction, the terminal proceeds to fee deduction by segment transaction, with transaction amount of 0.
- Based on its particular circumstances, the terminal, in UPDATE CAPP DATA CACHE, updates the subway toll custom data, fills in the city code, operator code, record format version number, transaction label, toll zone entry transaction time, toll zone entry line code, toll zone entry transaction station code, toll zone entry transaction gate code, toll zone entry transaction serial number, custom TAC, and other fields, and save the toll zone exit transaction time, toll zone exit transaction line code, toll zone exit transaction station code, toll zone exit transaction gate code, toll zone exit transaction amount, toll zone exit transaction serial number, and other original record values.
- At the end of the transaction, the terminal applies dynamic data authentication on the card based on the data returned by the card during the transaction process. Only after the card passes authentication would the terminal permit the cardholder to enter the toll zone.

E.1.1.2 Gate-exit transaction flow

Description

Gate-exit transaction has the basic flow: Select the PPSE payment environment, then select the qUICS application, read the extended application custom file, determine whether file content is correct, if so, based on gate-entry data, calculate the toll amount. Then proceed to deduct the toll, and update the extended application custom file, indicate normal transaction completion, while also prompting the cardholder to exit the toll zone.

Flow Chart

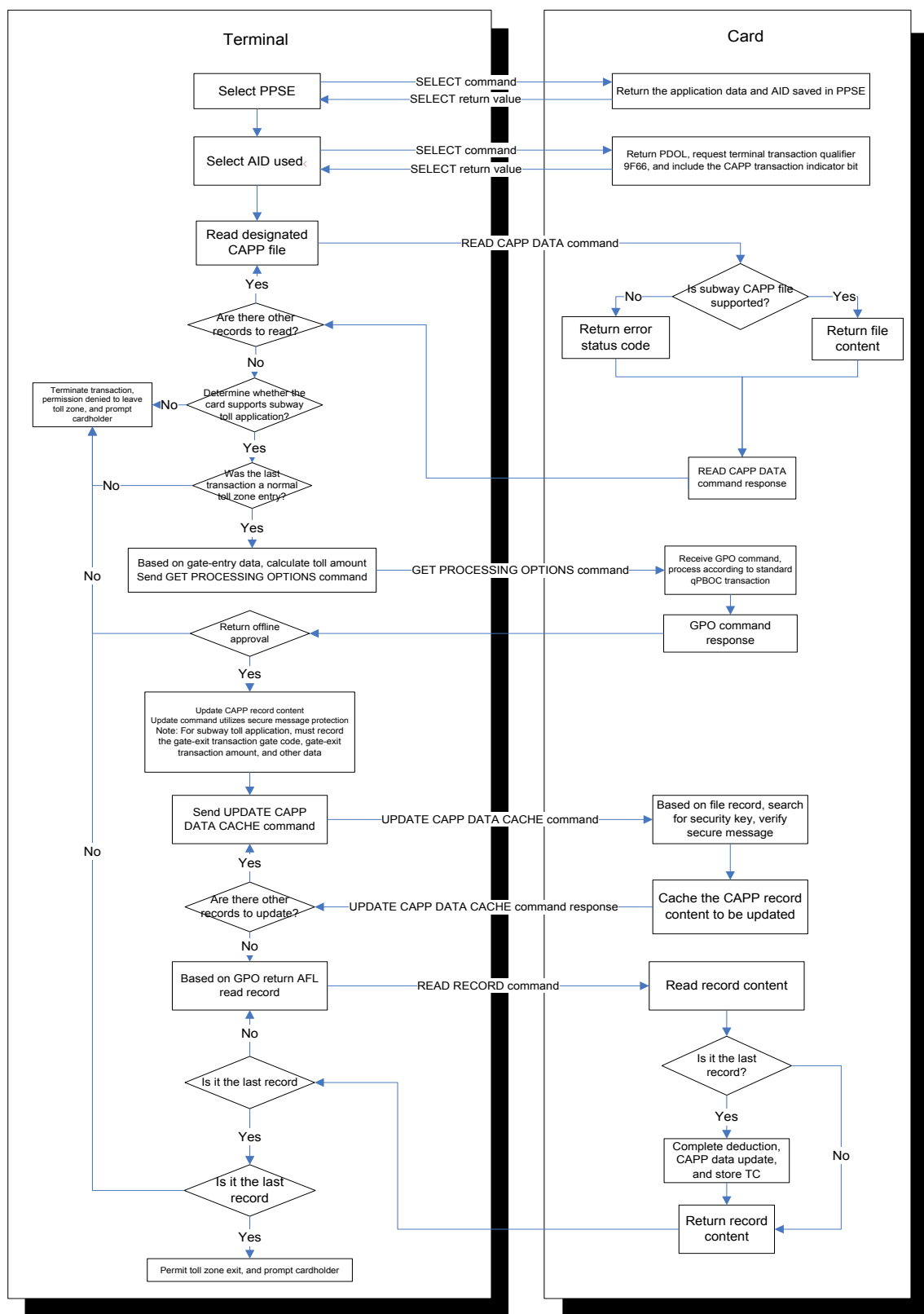


Figure E.2 Gate-exit transaction flow for subway toll application

Flow Description

When the cardholder uses contactless financial IC card for gate-exit transactions in a subway toll application environment, the terminal will process it as follows:

- Terminal first selects and activates the card, then through response data determine whether the card supports extended application transactions based on contactless micro-payments through AID selection.
- Terminal sends READ CAPP RECORD command inquiry, to determine whether the card supports subway toll application. If supported, the terminal shall read the subway toll custom data, and process according to the data, to determine whether the last transaction was a normal toll zone entry, etc, if so, then calculate the toll amount based on the gate-entry data in the extended application custom file. If the processing result is that the gate-exit transaction is not permitted, then the terminal shall notify the cardholder. If the processing result is that toll zone exit transaction is permitted, the terminal proceeds with deduction by segment transaction, and updates the extended application custom file, with the transaction amount being the toll amount calculated.
- Based on its particular circumstances, the terminal, in UPDATE CAPP DATA CACHE, updates the subway toll custom data, fills in the toll zone exit transaction time, toll zone exit line code, toll zone exit transaction station code, toll zone exit transaction gate code, toll zone exit transaction amount, toll zone exit transaction serial number, custom TAC, and other fields, and saves the city code, operator code, record format version number, transaction label, toll zone entry transaction time, toll zone entry transaction line code, toll zone entry transaction station code, toll zone entry transaction gate code, toll zone entry transaction serial number, and other original record values.
- At the end of the transaction, the terminal applies dynamic data authentication on the card based on the data returned by the card during the transaction process. Only after the card passes authentication would the terminal permit the cardholder to exit the toll zone.

E.1.2 Inter-city rail/Expressway Toll Applications

Inter-city rail toll / expressway toll applications use the same transaction flow as subway toll applications, also dividing the transaction into gate-entry transaction and gate-exit transaction, with the only difference being the toll standard. Each of them charges based on route segment, so at the time of gate-entry transaction, the recording of gate-entry transaction gate code and gate-entry transaction line code is very important, as these data are the basis for calculating toll amount at gate-exit.

E.1.3 Parking Meter Application

Parking meter application transaction flow is the same as that of the subway toll application, with the transaction divided into parking transaction and toll collection

transaction, equivalent to gate-entry transaction and gate-exit transaction. But subway toll application bills based on the segment ridden by the passenger, while the parking meter application bills based on the amount of parking time for the customer. So, for parking meter applications, at the time of the parking transaction, the recording of the transaction meter code and the parking transaction time is very important, as these data are the basis for calculating toll amount at the time of the toll collection transaction.

E.2 Public Transportation Day-Pass / Monthly-Pass Transaction Flow Description

Besides transactions that deduct by time or by segment, extended application can also realize public transportation day-pass / monthly-pass application functionality through reading the extended application custom file.

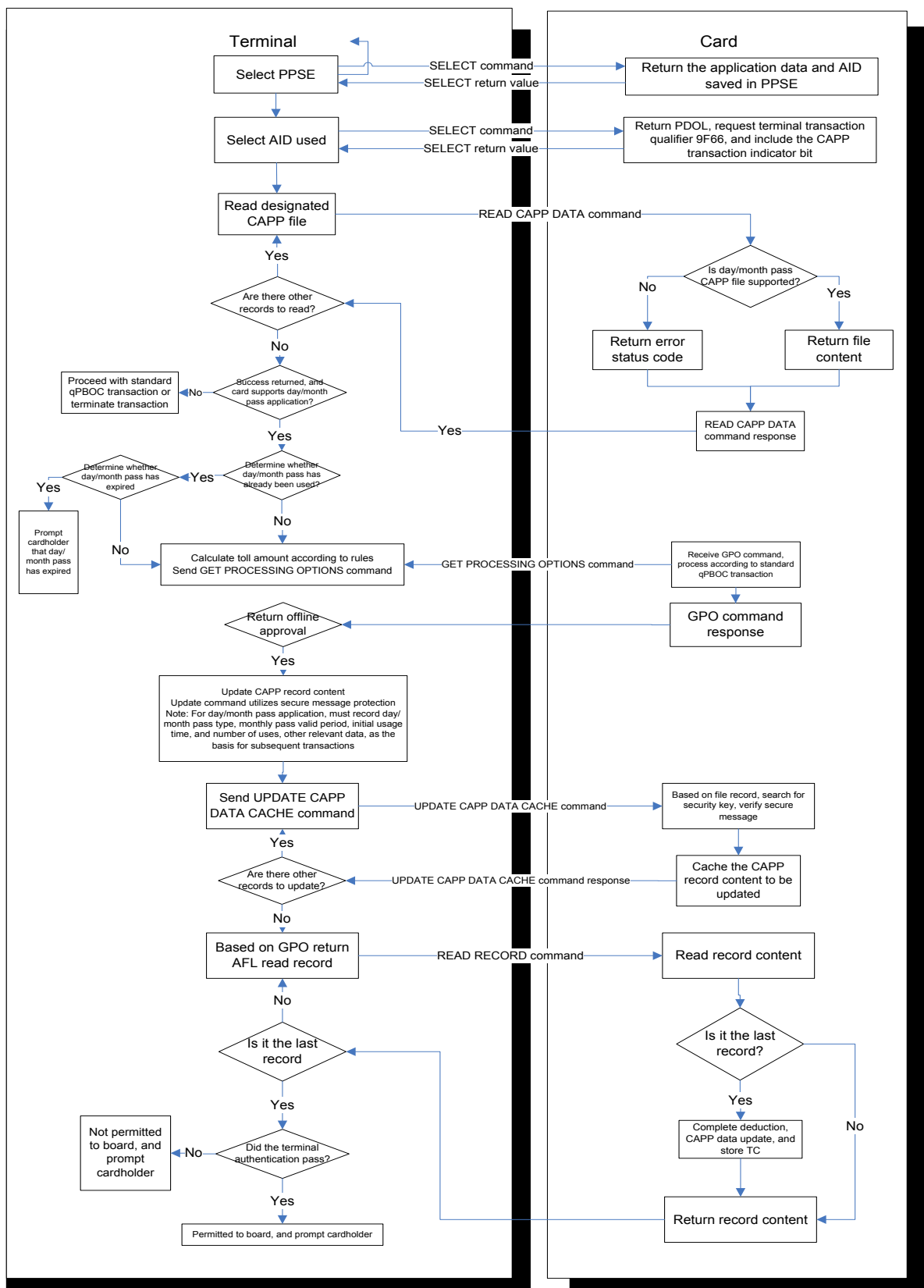
E.2.1 Public Transportation Day-Pass / Monthly-Pass Application

Description

Extended applications include the following two types in the public transportation day/month pass domain: limited-use type and unlimited-use type. Whereby, limited-use type means limiting the number of uses of the day/month pass during the day/month, engaging in equal amount tolls, with the toll amount being the proportional value of the day/month pass total amount and the limited number of uses; unlimited-use type means there is no limit to the number of uses for the day/month pass during the day/month, and deducts the entire day/month amount on the first use, thereafter engaging in 0 amount tolls on subsequent uses.

The basic flow for the day/month pass transaction is: Read the extended application custom file, determine whether the card supports public transportation day/month pass application, if yes, determine whether the public transportation day/month pass has been used already; if not yet used, then proceed to day/month pass toll transaction; if already used, then based on initial usage time and (/or) number of uses, determine whether the day/month pass has already expired, if expired, then notify the cardholder that the day/month pass has expired, otherwise continue with day/month pass toll transaction.

Flow Chart



Public transportation day/month pass toll transaction flow chart

Flow Description

When the cardholder uses contactless financial IC card for public transportation day/month pass transactions in a day/month pass application environment, the terminal will process it as follows:

- Terminal first selects and activates the card, then through response data determine whether the card supports extended application transactions based on contactless micro-payments.
- Terminal sends READ CAPP RECORD command to inquire the industry file, to determine whether the card supports public transportation day/month pass application. If supported, the terminal shall read the public transportation day/month pass custom data, then process based on the data: first determine whether the public transport day/month pass has already been used: if not used, then based on the rules, calculate the toll amount and proceed with day/month pass toll transaction; if already used, then based on the initial usage time and (/or) number of uses, determine whether the day/month pass has expired, if expired, then terminate the transaction and notify the cardholder that the day/month pass has expired, if not expired then calculate the toll amount based on the rules and continue with day/month pass toll transaction.
- For day/month pass applications, the extended application custom file must record the day/month pass type, day/month pass effective period, initial usage time, number of times used, and other relevant data, as the basis for subsequent transactions.
- If the day/month pass must be used on specific day/month, then the CAPP file can be updated during card top-up/issuance.

Appendix F (Informative Annex) Industry Application Setup Guide

Setting up an industry application master key is generally managed by the issuer, and each industry application is controlled by independent industry application setup master keys, so as to ensure the independence of each industry. For IC card application setup key derivation method, see the description for sub-key derivation in UICS, with industry application setup master key derived with financial application PAN number and PAN serial number.

Industry application management and setup process is as follows:

- Issuer generates the industry application setup master key with its IC card key management system.
- When the issuer is preparing the IC card data, the industry application setup master key is derived with the financial application PAN number and PAN serial number to obtain the IC card industry application setup key.
- During issuer personalization, extended application file is preemptively created, setting up the corresponding IC card industry application setup key.
- The cardholder, on the designated terminal and under the protection of the industry application setup key, adds industry application record through the APPEND RECORD command to setup the industry application.

Industry applications can be setup through the following route:

1) Terminal has the means of authentication to setup the industry application: Terminal contains the industry application setup master key, through PAN number and PAN serial number derivation, obtains the IC card industry application setup key. The terminal, under the control of the IC card industry application setup key, creates the industry application record (industry application management key is generated by the issuer or industry cooperative, encrypted using the IC card industry application setup key, and written into the IC card).

2) Use issuer backend authentication means to setup industry application: The terminal does not contain the industry application setup master key, but rather the industry application setup master key is stored on the issuer backend, with the card and issuer backend engaging in online interactive authentication, with the same application flow for industry application setup as for the terminals with the means of authentication. This method is suitable for remote setup of industry applications.

Appendix G New IDD issuer discretionary data (9F10)

For issuer discretionary data (9F10) in contactless IC card micro-payment extended application's offline pre-authorization, add data with IDD ID of 0x06, with data adding new IDD issuer discretionary data as in Table G.1 below:

Table G.1 new IDD issuer discretionary data

Issuer discretionary data option	Length (byte)	IDD ID	Financial Field	MAC Byte Size
Authorized Transaction Amount	10	0x06	Tag "9F02" value (lower 5 bytes)	4

For offline pre-authorization completion and deduction by segment transactions with issuer discretionary data (9F10) from original qUICS, contactless IC card micro-payment extend applications, use the original IDD ID01~05 data.

Through the newly added IDD ID, the issuer can undertake special processing for not-yet-completed offline pre-authorization transactions. This processing requires support from the corresponding issuer backend system.